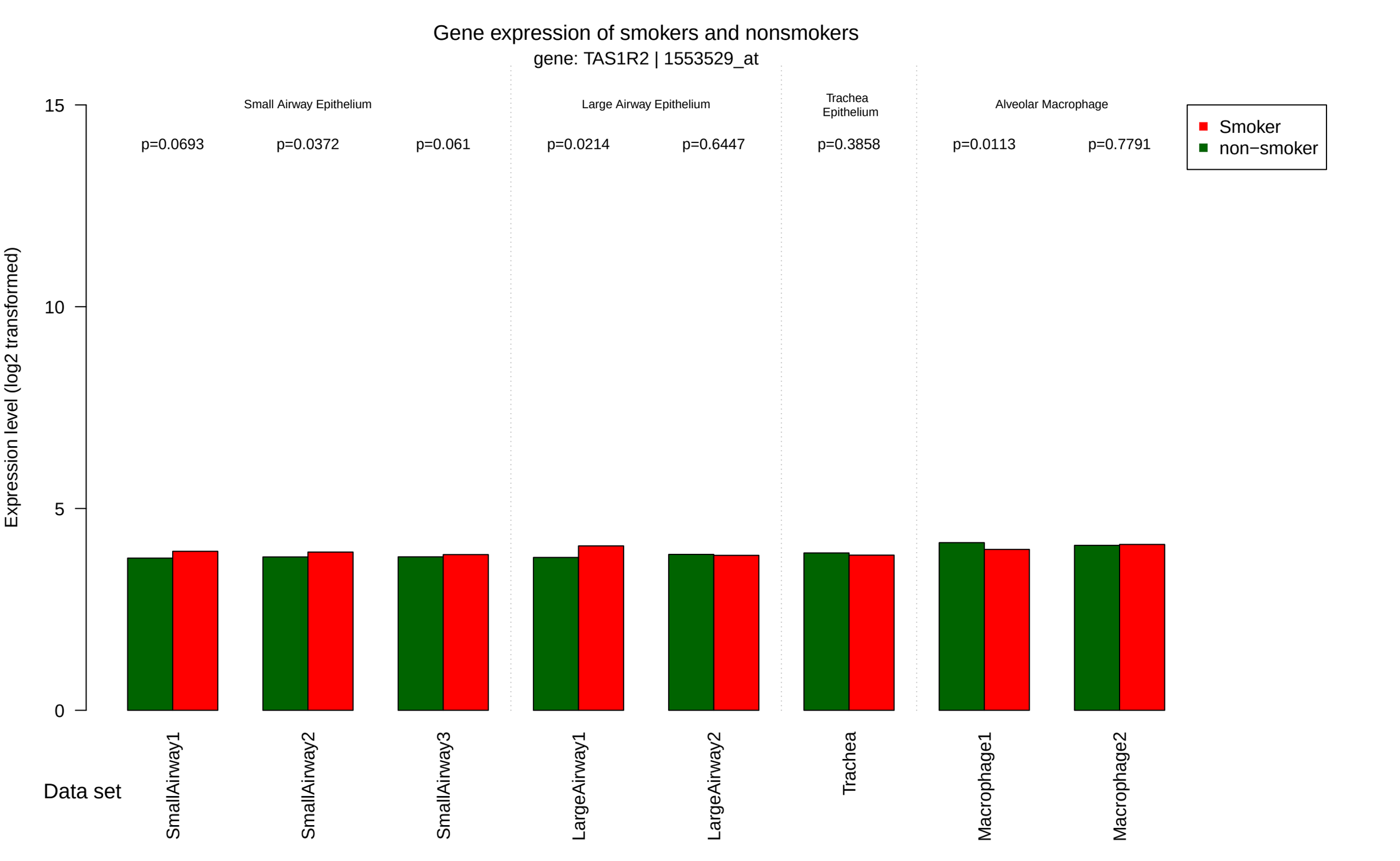
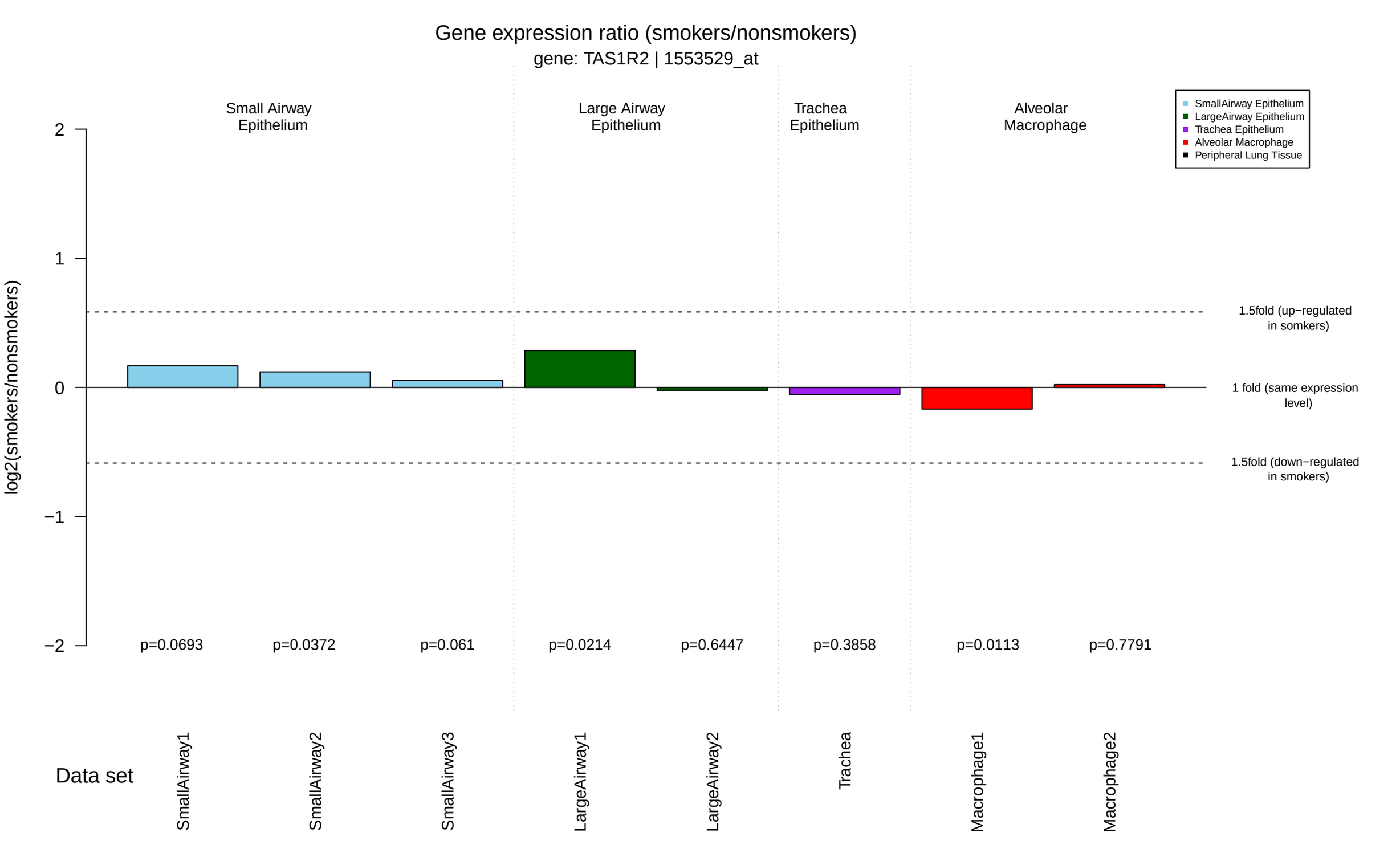
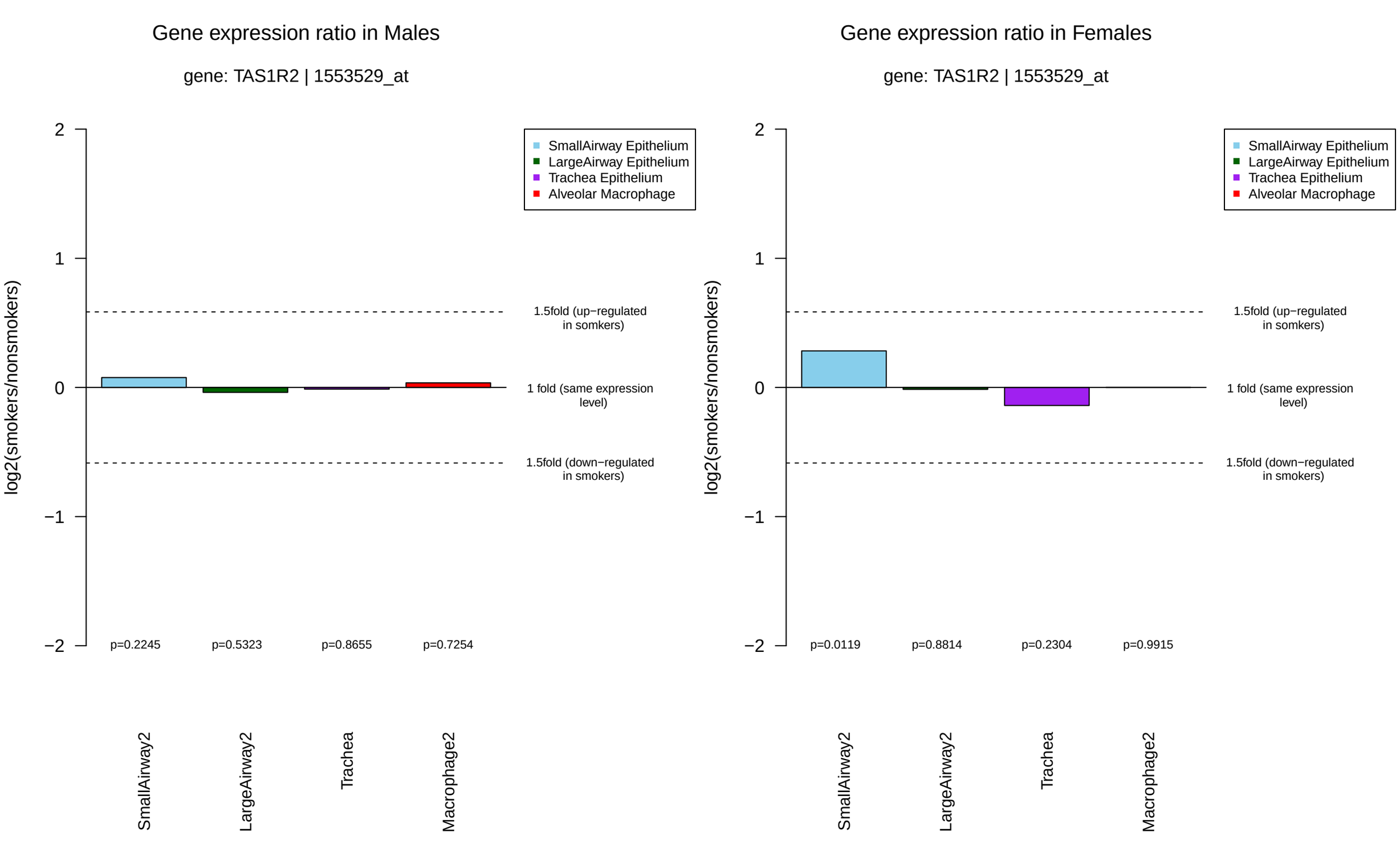
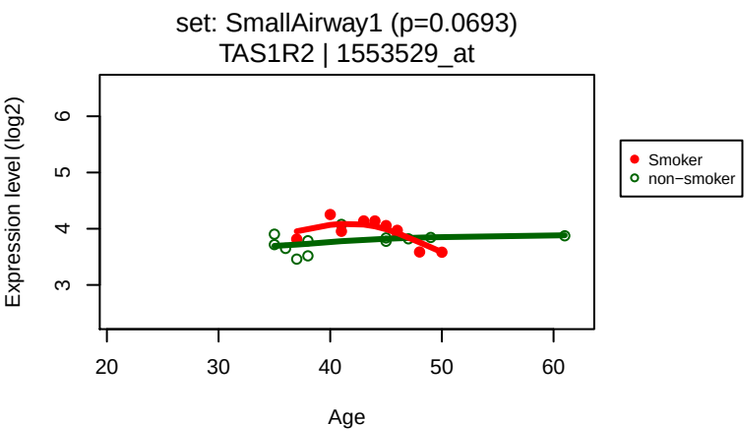
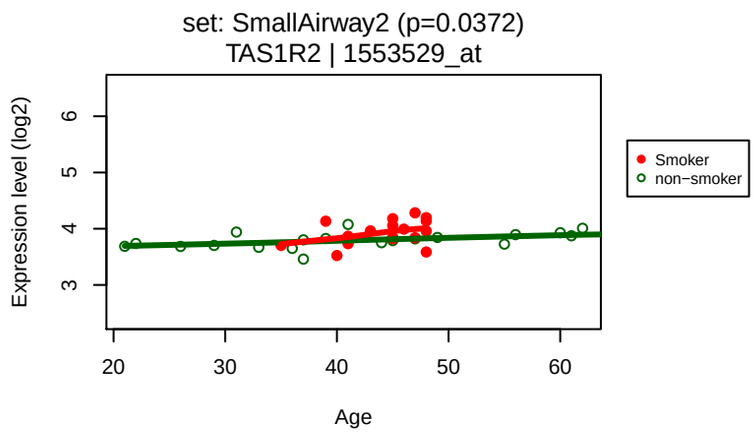
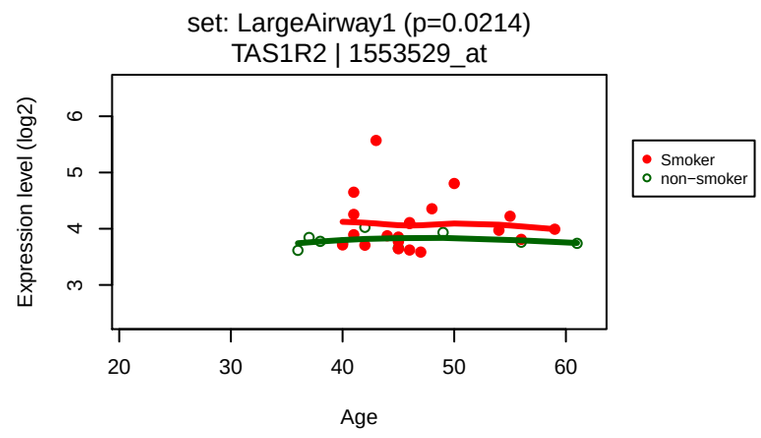

Gene expression of smokers and nonsmokers
gene: TAS1R2 | 1553529_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0693
p=0.0372
p=0.061
p=0.0214
p=0.6447
p=0.3858
p=0.0113
p=0.7791
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

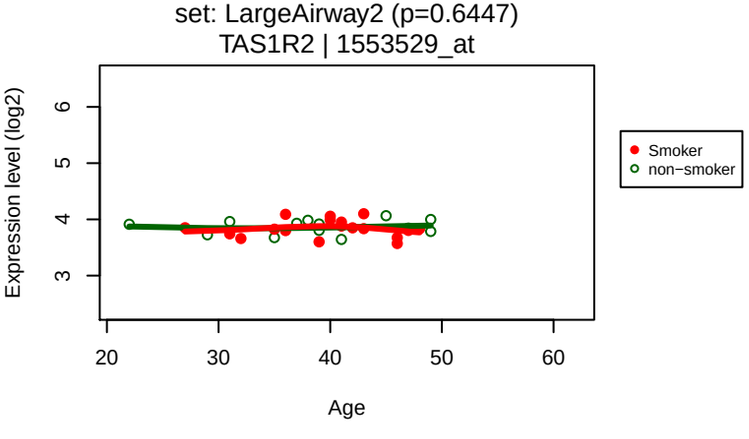

Gene expression ratio (smokers/nonsmokers)
gene: TAS1R2 | 1553529_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0693
p=0.0372
p=0.061
p=0.0214
p=0.6447
p=0.3858
p=0.0113
p=0.7791
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

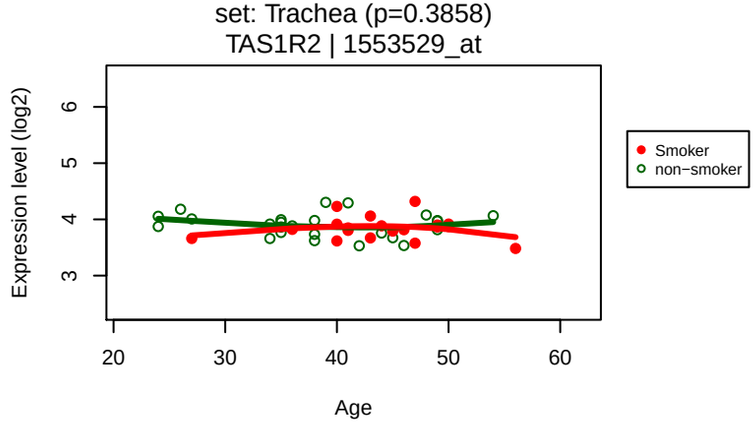

Gene expression ratio in Males
gene: TAS1R2 | 1553529_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2245
p=0.5323
p=0.8655
p=0.7254
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TAS1R2 | 1553529_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0119
p=0.8814
p=0.2304
p=0.9915
SmallAirway2
LargeAirway2
Trachea
Macrophage2

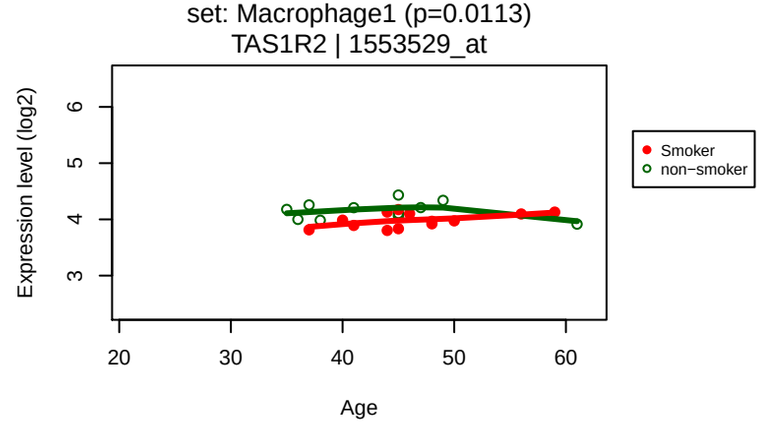

set: SmallAirway1 (p=0.0693)
TAS1R2 | 1553529_at
Expression level (log2)
Age
Smoker
non−smoker

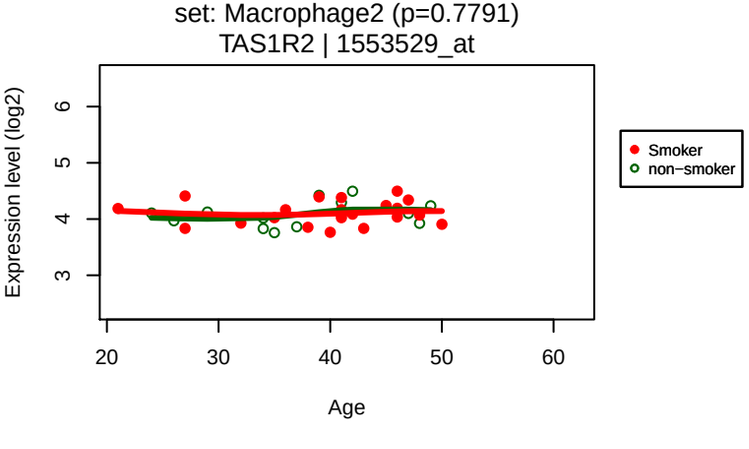

set: SmallAirway2 (p=0.0372)
TAS1R2 | 1553529_at
Expression level (log2)
Age
Smoker
non−smoker

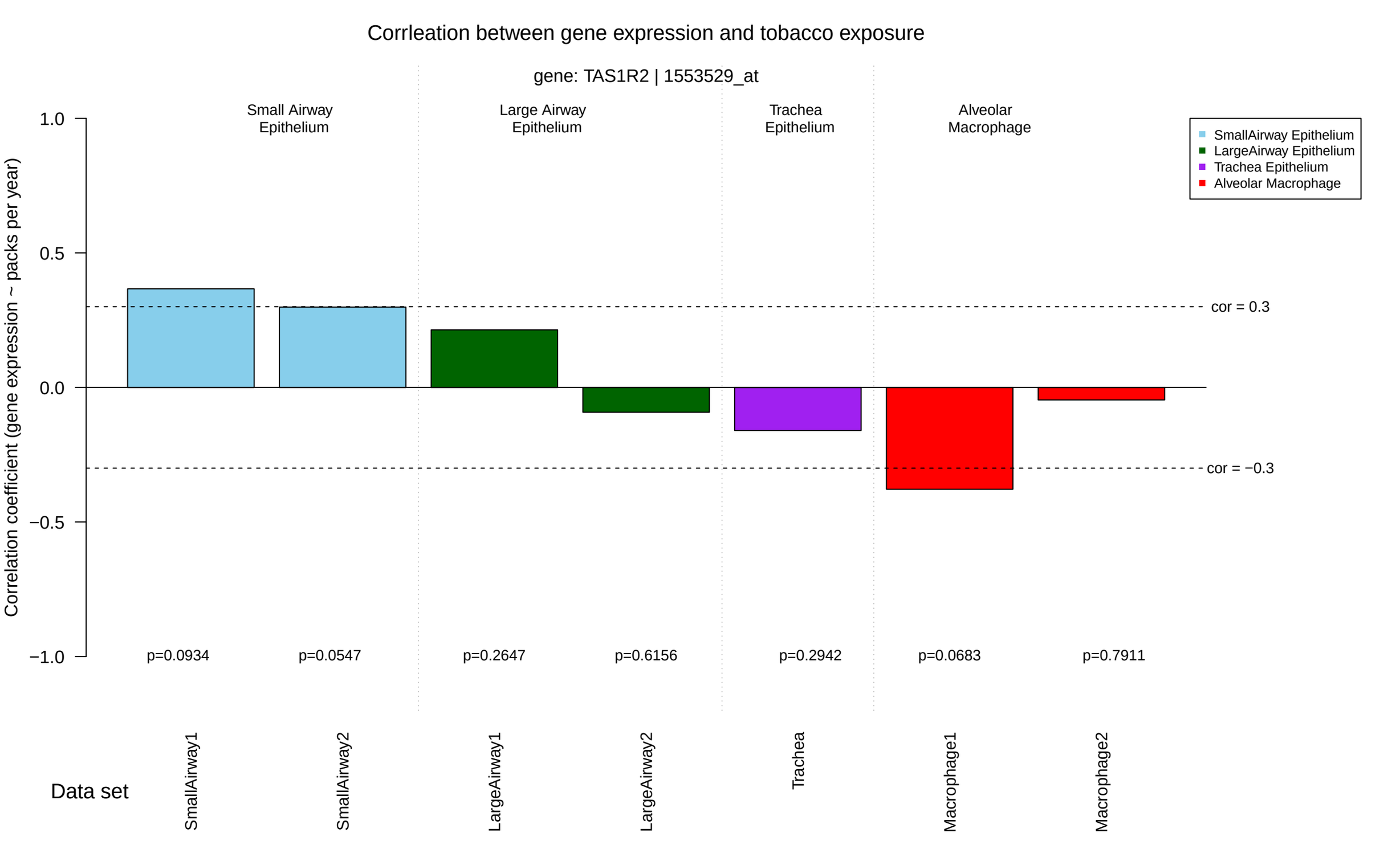

set: LargeAirway1 (p=0.0214)
TAS1R2 | 1553529_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6447)
TAS1R2 | 1553529_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3858)
TAS1R2 | 1553529_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0113)
TAS1R2 | 1553529_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7791)
TAS1R2 | 1553529_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TAS1R2 | 1553529_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0934
p=0.0547
p=0.2647
p=0.6156
p=0.2942
p=0.0683
p=0.7911
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2